

PROJECT DOCUMENTATION

Case Study

Arrays

Lotto - A lottery program made of 6 random numbers (Lotto 6/42).

Design a program that:

- Allows the user to enter 6 numbers as their bet.
- Generates the 6 random numbers for the winning combination. (Use the random function in Java.)
- Display the winning numbers to make it easier to check.
- Display the prize of the lotto buyer based on the following:

Winning Numbers (in any order)	Prize
6 winning numbers	P 31,621,223.97
5 winning numbers	P 25,000.00
4 winning numbers	P 1,000.00
3 winning numbers	P 20.00

The screenshot shows an IDE window titled 'Mavenproject3 - Apache NetBeans IDE 28'. The left sidebar displays the project structure with 'Mavenproject3' expanded, showing 'Source Packages' and 'Mavenproject3.java'. The main editor area shows the source code of 'Mavenproject3.java'. The code includes a package declaration, imports for 'InputMismatchException', 'Scanner', and 'Random', and a 'main' method. The 'main' method prompts the user to enter 6 numbers, validates them to be between 1 and 42, and generates 6 random numbers for the winning combination. The code is as follows:

```
1  /* Click https://nhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license */
2
3
4  package com.mycompany.mavenproject3;
5
6
7  /**
8   *
9   * @author Andrew San Antonio
10  */
11
12  import java.util.InputMismatchException;
13  import java.util.Scanner;
14  import java.util.Random;
15
16  public class Mavenproject3 {
17
18      public static void main(String[] args) {
19          Scanner scanner = new Scanner(System.in);
20          int[] playerNumber = new int[6];
21          int[] lottoNumber = generateNumber();
22
23          System.out.println("--- Lotto ---");
24          // User enter 6 number
25          for (int i = 0; i < playerNumber.length; i++) {
26              while (true) {
27                  try {
28                      System.out.print("Enter number #" + (i + 1) + " : ");
29                      playerNumber[i] = scanner.nextInt();
30
31                      if (playerNumber[i] > 0 && playerNumber[i] <= 42) {
32                          break; // valid input
33                      } else {
34                          System.out.println("Number must be between 1 and 42!");
35                      }
36                  } catch (InputMismatchException e) {
37                      System.out.println("Please enter a number!");
38                      scanner.next();
39                  }
40              }
41          }
42      }
43  }
```

```
38 }
39
40 // New Line
41 System.out.println();
42
43 // Display Bet Numbers and Lotto Numbers
44 displayNumbers(playerNumber, lottoNumber);
45 // Display Winning Numbers
46 displayWinningNumbers(playerNumber, lottoNumber);
47 // Display Prize of the winners
48 displayPrize(playerNumber, lottoNumber);
49 }
50 // Generate 6 random numbers
51 public static int[] generateNumber() {
52     Random random = new Random();
53
54     int[] lottoNumber = new int[6];
55
56     for (int i = 0; i < lottoNumber.length; i++) {
57         lottoNumber[i] = random.nextInt(42) + 1;
58     }
59
60     return lottoNumber;
61 }
62 // display player winning numbers
63 public static void displayWinningNumbers(int[] playerNumber, int[] lottoNumber) {
64     System.out.println("Winning numbers");
65     boolean isMatch = false;
66
67     for (int i = 0; i < lottoNumber.length; i++) {
68         for (int j = 0; j < playerNumber.length; j++) {
69             if (lottoNumber[i] == playerNumber[j]) {
70                 System.out.print(playerNumber[j] + " ");
71                 isMatch = true;
72             }
73         }
74     }
75 }
```

```
75
76 if (!isMatch) {
77     System.out.println("Not found!");
78 }
79 System.out.println();
80 public static void displayNumbers(int[] playerNumber, int[] lottoNumber) {
81     System.out.println("Bet Numbers:");
82     for (int i : playerNumber) {
83         System.out.print(i + " ");
84     }
85     System.out.println();
86     System.out.println("Generated Numbers:");
87     for (int i : lottoNumber) {
88         System.out.print(i + " ");
89     }
90     System.out.println();
91 }
92 // Display prize of the winners
93 public static void displayPrize(int[] playerNumber, int[] lottoNumber) {
94     int count = 0;
95     double prize = 0;
96
97     // Nested Loop
98     for (int i = 0; i < lottoNumber.length; i++) {
99         for (int j = 0; j < playerNumber.length; j++) {
100             // Check Matching numbers
101             if (lottoNumber[i] == playerNumber[j]) {
102                 // If True count will add 1
103                 count++;
104             }
105         }
106     }
107 }
```

```
107 // Prize
108 switch (count) {
109     case 6 -> prize = 31621223.97;
110     case 5 -> prize = 25000;
111     case 4 -> prize = 1000;
112     case 3 -> prize = 20;
113     default -> {
114     }
115 }
116 System.out.println("\nYou won: ₱ " + String.format("%.2f", prize));
117 }
118 }
119 }
```

Sample Output 1

```
--- Lotto ---  
Enter number #1 : 25  
Enter number #2 : 34  
Enter number #3 : 12  
Enter number #4 : R  
Please enter a number!  
Enter number #4 : 45  
Number must be between 1 and 42!  
Enter number #4 : 21  
Enter number #5 : 12  
Enter number #6 : 32
```

```
Bet Numbers:  
25 34 12 21 12 32  
Generated Numbers:  
11 18 11 14 33 9  
Winning numbers  
Not found!
```

```
You won: ? 0.00
```

```
-----  
BUILD SUCCESS  
-----
```

Methods

```

11 import java.util.Scanner;
12 public class Mavenproject4
13 {
14     public static void main(String[] args) {
15         Scanner scanner = new Scanner(System.in);
16
17         System.out.print("Enter your height (in cm): ");
18         double height = scanner.nextDouble();
19         System.out.print("Enter your weight (in kilograms): ");
20         double weight = scanner.nextDouble();
21
22         calculateBMI(height, weight);
23
24     }
25     // Calculate BMI
26     public static void calculateBMI(double height, double weight) {
27         double bmi = 0;
28         String category = "";
29         bmi = (weight / (height * height)) * 10000;
30
31         if (bmi >= 30) {
32             category = "Obesity";
33         } else if (bmi > 25) {
34             category = "Overweight";
35         } else if (bmi > 18.5) {
36             category = "Normal weight";
37         } else {
38             category = "Underweight";
39         }
40         System.out.printf("BMI: %.2f\n", bmi);
41         System.out.println("Category: " + category);
42     }
43 }
44

```

Sample Output 1

```

Enter your height (in cm): 120
Enter your weight (in kilograms): 50
BMI: 34.72
Category: Obesity

```

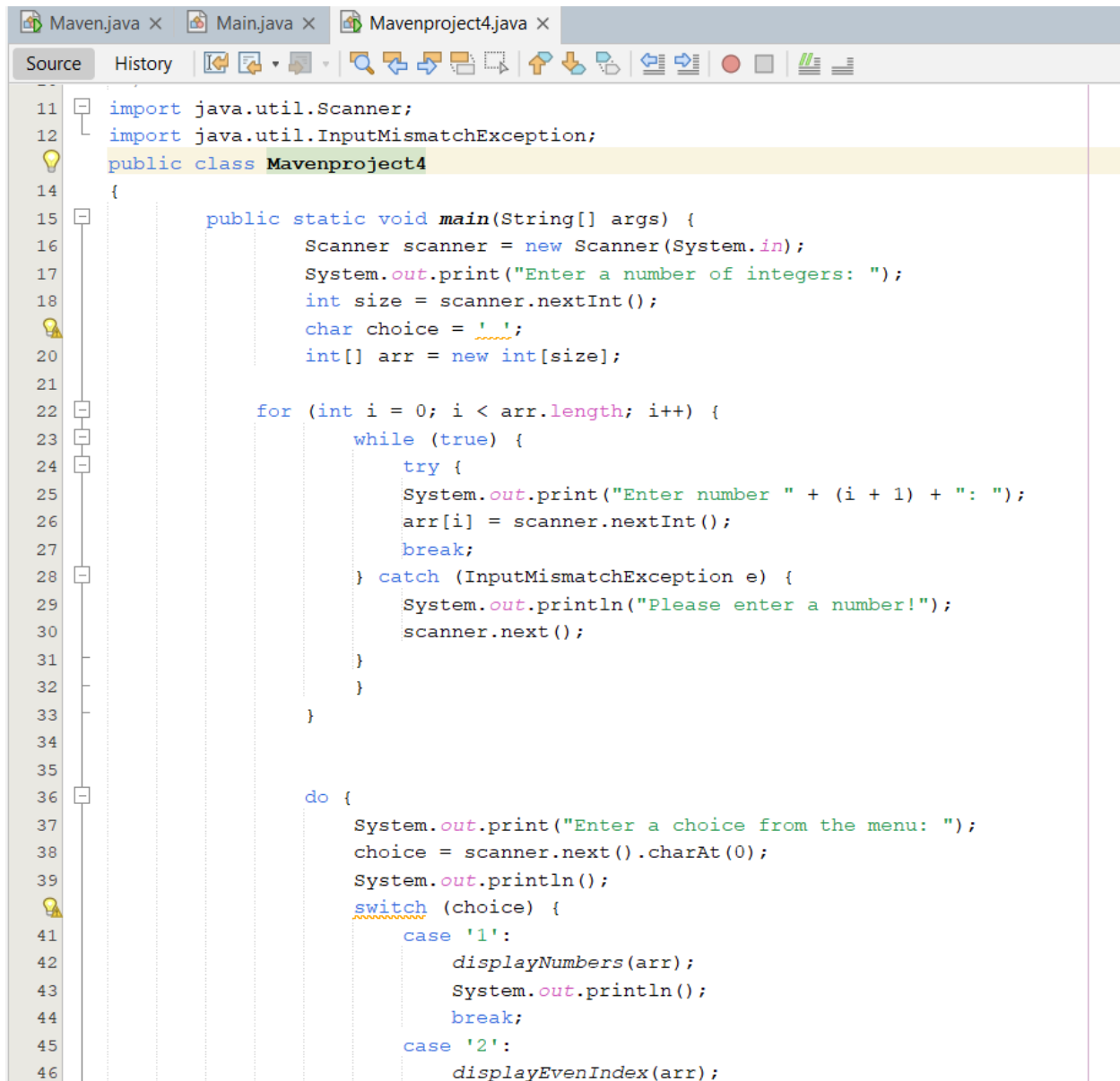
BUILD SUCCESS

```

Total time: 33.918 s
Finished at: 2026-05-02T13:37:48+08:00

```

Sample Output 2



```
11 import java.util.Scanner;
12 import java.util.InputMismatchException;
13 public class Mavenproject4
14 {
15     public static void main(String[] args) {
16         Scanner scanner = new Scanner(System.in);
17         System.out.print("Enter a number of integers: ");
18         int size = scanner.nextInt();
19         char choice = ' ';
20         int[] arr = new int[size];
21
22         for (int i = 0; i < arr.length; i++) {
23             while (true) {
24                 try {
25                     System.out.print("Enter number " + (i + 1) + ": ");
26                     arr[i] = scanner.nextInt();
27                     break;
28                 } catch (InputMismatchException e) {
29                     System.out.println("Please enter a number!");
30                     scanner.next();
31                 }
32             }
33         }
34
35         do {
36             System.out.print("Enter a choice from the menu: ");
37             choice = scanner.next().charAt(0);
38             System.out.println();
39             switch (choice) {
40                 case '1':
41                     displayNumbers(arr);
42                     System.out.println();
43                     break;
44                 case '2':
45                     displayEvenIndex(arr);
46             }
47         } while (choice != '0');
```

```
47         System.out.println();
48         break;
49         case '3':
50             displayOddIndex(arr);
51             System.out.println();
52             break;
53         case '4':
54             displayAscOrder(arr);
55             System.out.println();
56             break;
57         case '5':
58             displayDescOrder(arr);
59             System.out.println();
60             break;
61     }
62     } while (choice != 'X');
63 }
64 public static void displayNumbers(int[] arr) {
65     System.out.print("Numbers: ");
66     for (int i = 0; i < arr.length; i++) {
67         System.out.print(arr[i] + " ");
68     }
69     System.out.println();
70 }
71 public static void displayEvenIndex(int[] arr) {
72     System.out.print("Values in even indexes: ");
73     for (int i = 0; i < arr.length; i++) {
74         if (i % 2 == 0) {
75             System.out.print(arr[i] + " ");
76         }
77     }
78     System.out.println();
79 }
80 public static void displayOddIndex(int[] arr) {
81     System.out.print("Values in odd indexes: ");
82     for (int i = 0; i < arr.length; i++) {
83         if (i % 2 != 0) {
```

```

84         System.out.print(arr[i] + " ");
85     }
86 }
87 System.out.println();
88 }
89 public static void displayAscOrder(int[] arr) {
90     for (int i = 0; i < arr.length; i++) {
91         for (int j = 0; j < arr.length - i - 1; j++) {
92             if (arr[j] > arr[j + 1]) {
93                 int temp = arr[j];
94                 arr[j] = arr[j + 1];
95                 arr[j + 1] = temp;
96             }
97         }
98     }
99     System.out.print("Values in ascending order: ");
100     for (int i : arr) {
101         System.out.print(i + " ");
102     }
103     System.out.println();
104 }
105 public static void displayDescOrder(int[] arr) {
106     for (int i = 0; i < arr.length; i++) {
107         for (int j = 0; j < arr.length - i - 1; j++) {
108             if (arr[j] > arr[j + 1]) {
109                 int temp = arr[j];
110                 arr[j] = arr[j + 1];
111                 arr[j + 1] = temp;
112             }
113         }
114     }
115     System.out.print("Values in descending order: ");
116     for (int i = arr.length - 1; i >= 0; i--) {
117         System.out.print(arr[i] + " ");
118     }
119     System.out.println();
120 }

```

Sample Output 1


```
--- exec:3.5.1:exec (default-cli) @ mavenproject4 ---
Enter a number of integers: 6
Enter number 1: 50
Enter number 2: 62
Enter number 3: 123
Enter number 4: 12354
Enter number 5: 3654
Enter number 6: 121
Enter a choice from the menu: 4

Values in ascending order: 50 62 121 123 3654 12354

Enter a choice from the menu: X
```

BUILD SUCCESS

Total time: 01:30 min
Finished at: 2026-05-02T13:48:38+08:00

Stacks and Queues

```

11 import java.util.Scanner;
12 public class Mavenproject4
13 {
14     public static void main(String[] args) {
15         Scanner scanner = new Scanner(System.in);
16         String data = "";
17         System.out.print("Enter any text: ");
18         data = scanner.nextLine();
19
20         char[] ch = data.toCharArray();
21         int size = ch.length;
22
23         char choice = ' ';
24         do {
25             System.out.print("Enter a choice from the menu [Press X to exit the program]: ");
26             choice = scanner.next().charAt(0);
27
28             switch(choice) {
29                 case 'A':
30                     for (int i = 0; i < size; i++) {
31                         System.out.print(ch[i]);
32                     }
33                     System.out.println();
34                     break;
35                 case 'B':
36                     if (size == 0) {
37                         System.out.println("Empty");
38                     } else {
39                         System.out.println("Character Popped");
40                         size--;
41                     }
42                     break;
43                 case 'C':
44                     if (size == 0) {
45                         System.out.println("Empty");
46                     } else {
47                         System.out.println("Top character is " + ch[size - 1]);

```

```

48     }
49     break;
50 case 'D':
51     System.out.print("Enter character: ");
52     char letter = scanner.next().charAt(0);
53
54     char[] newLetter = new char[size + 1];
55     for (int i = 0; i < size; i++) {
56         newLetter[i] = ch[i];
57     }
58     newLetter[size] = letter;
59     ch = newLetter;
60     size++;
61     System.out.println("Character Pushed/Enqueued");
62     break;
63 case 'E':
64     if (size == 0) {
65         System.out.println("Empty");
66     } else {
67         for (int i = 1; i < size; i++) {
68             ch[i - 1] = ch[i];
69         }
70         size--;
71         for (int i = 0; i < size; i++) {
72             System.out.print(ch[i]);
73         }
74     }
75     System.out.println();
76     break;
77 case 'F':
78     if (size == 0) {
79         System.out.println("Empty");
80     } else {
81         System.out.println("Front Character is " + ch[0]);
82     }
83     break;
84 case 'X':
85     System.out.println("End");
86     break;
87 }
88 } while (choice != 'X');
89 }
90 }
91
92

```

Sample Output 1

Enter any text: PERSEVERE

- A. Display the string.
- B. Pop
- C. Top
- D. Pushed/Enqueued
- E. Dequeue
- F. Front

Enter a choice from the menu [Press X to exit the program]: A
PERSEVERE

Enter a choice from the menu [Press X to exit the program]: D
Enter character: S

Character Pushed/Enqueued

Enter a choice from the menu [Press X to exit the program]: A
PERSEVERES

Enter a choice from the menu [Press X to exit the program]: E
ERSEVERES

Enter a choice from the menu [Press X to exit the program]: F
Front Character is E

Enter a choice from the menu [Press X to exit the program]: B
Character Popped

Enter a choice from the menu [Press X to exit the program]: A
ERSEVERE

Enter a choice from the menu [Press X to exit the program]: X
End

BUILD SUCCESS
